

ISO 26262: Road Vehicles – Functional Safety

Comprehensive Overview for ADAS & Autonomous Driving Engineers

Knowledge Base Document · RAG-Ready · ADAS Domain

1. Introduction and Scope

ISO 26262, titled 'Road Vehicles – Functional Safety', is the primary international standard governing the development of safety-related electrical and electronic (E/E) systems in road vehicles. First published in 2011 as an adaptation of IEC 61508 (the generic functional safety standard), its second edition was released in December 2018, extending coverage from passenger cars to all road vehicles except mopeds.

The standard addresses hazards caused by the malfunctioning behaviour of E/E systems. It provides a framework for identifying, classifying, and mitigating safety risks throughout the entire product lifecycle — from concept through decommissioning. ISO 26262 is widely adopted by OEMs and Tier 1 suppliers as a de facto requirement for ADAS, powertrain, chassis, and body electronics development.

1.1 Objectives of ISO 26262

- Provide a reference framework for automotive functional safety activities.
- Define specific requirements for software, hardware, and system-level development at each ASIL level.
- Ensure that safety risks from E/E system failures are reduced to an acceptable level.
- Facilitate communication of safety requirements between OEMs, Tier 1 and Tier 2 suppliers.
- Support functional safety management throughout the supply chain and development lifecycle.

1.2 Relationship to Other Standards

ISO 26262 is part of a family of automotive safety and quality standards. ISO 26262 focuses on functional safety (freedom from unreasonable risk due to E/E malfunctions). It is complementary to — but distinct from — ISO/PAS 21448 (SOTIF: Safety of the Intended Functionality), which addresses hazards from the correct behaviour of a system in situations where it was not designed to handle, particularly relevant for AI-based perception systems. ISO 21434 governs cybersecurity for road vehicles. AUTOSAR provides the software architecture framework within which ISO 26262-compliant software is developed.

2. Automotive Safety Integrity Levels (ASIL)

The Automotive Safety Integrity Level (ASIL) is the central risk classification mechanism in ISO 26262. It is determined during the Hazard Analysis and Risk Assessment (HARA) process and ranges from QM (Quality Management, no specific safety requirements) to ASIL D (the highest integrity level). The ASIL quantifies the safety risk of a hazardous event and drives the stringency of all subsequent safety requirements.

ASIL Level	Risk Level	Typical Application	Required Measures
QM (Quality Mgt)	No safety requirement	Non-safety-relevant functions	Standard quality management only
ASIL A	Lowest safety integrity	Windscreen wiper, interior lighting	Basic SW tests, code reviews
ASIL B	Low-medium integrity	ACC (highway), power windows	FMEA, unit testing, fault detection

ASIL C	Medium-high integrity	AEB (partial), EPS assistance	Formal analysis, MC/DC coverage, FMEDA
ASIL D	Highest safety integrity	AEB (full), brake-by-wire	Formal verification, hardware redundancy, ASIL D SW requirements

2.1 ASIL Determination Parameters

The ASIL of a safety goal is determined by evaluating three parameters for each hazardous event identified in the HARA:

- **Severity (S0–S3):** Potential harm to persons. S0 = no injuries; S1 = light injuries; S2 = severe/life-threatening injuries (survival probable); S3 = life-threatening injuries (survival uncertain) or fatal.
- **Exposure (E0–E4):** How frequently the vehicle and its occupants are in the operational situation where the hazardous event could occur. E0 = incredibly unlikely; E4 = high probability (many driving hours per day).
- **Controllability (C0–C3):** The ability of a typical driver or other person to avoid the harm. C0 = controllable in general; C3 = difficult or impossible to control.

ASIL is determined by a combination table: high severity + high exposure + low controllability → ASIL D. For example, an ACC false braking event on a motorway (S2, E4, C2) yields ASIL B. An AEB false activation at 100 km/h (S3, E3, C3) yields ASIL D.

2.2 ASIL Decomposition

ISO 26262 allows ASIL decomposition: a single ASIL D requirement can be split into two independent ASIL B requirements allocated to redundant subsystems (ASIL B + ASIL B = ASIL D, provided independence is demonstrated). This technique is widely used in ADAS safety architectures to achieve the required integrity through redundancy rather than implementing a single, extremely complex ASIL D component.

3. The ISO 26262 Safety Lifecycle

ISO 26262 defines a safety lifecycle covering all phases from concept through operation and decommissioning. Development activities follow a V-model structure, with requirements decomposition on the left arm and verification/validation activities on the right arm mirroring each phase.

3.1 Concept Phase (Part 3)

The concept phase is the most critical phase because it establishes the safety goals that drive all subsequent requirements. The key activities are:

- **Item Definition:** Define the system boundary, interfaces, operating modes, and assumptions about the operational environment. For ACC: define the radar/camera inputs, CAN interfaces, actuator outputs (brakes, throttle), and the SAE Level 2 automation boundary.
- **Hazard Analysis and Risk Assessment (HARA):** Systematically identify all hazardous events that could result from the item's malfunction. For each hazardous event, determine the operational situation, assess S/E/C parameters, and derive the ASIL. The output is a set of safety goals, each with an ASIL.
- **Functional Safety Concept (FSC):** Define the high-level safety measures to achieve the safety goals. Specifies which parts of the system are safety-relevant and at what ASIL.

3.2 System Development (Part 4)

The system level translates functional safety requirements into technical safety requirements and a system architecture:

- **Technical Safety Concept (TSC):** Refines the FSC into technical safety requirements at the system level, allocating ASIL requirements to hardware and software elements.
- **System Architectural Design:** Defines the hardware/software partitioning. Safety mechanisms such as watchdogs, plausibility checks, and redundant sensors are specified here.
- **Safety Analysis:** Systematic methods (FMEA, FTA, FMEDA) are applied to verify that safety mechanisms adequately prevent or mitigate identified failure modes.
- **Integration and Testing:** Hardware-software integration testing, system testing, and validation against the safety goals.

3.3 Hardware Development (Part 5)

- **Hardware Architectural Design:** Defines hardware components, their safety mechanisms, and diagnostic coverage.
- **PMHF (Probabilistic Metric for Hardware Failures):** Quantitative target for random hardware failures. ASIL D requires $PMHF < 10^{-8}$ per hour of operation.
- **SPFM / LFM:** Single-Point Fault Metric and Latent Fault Metric — quantitative measures of how well the hardware architecture covers single-point and latent failures.
- **Hardware Integration Testing:** Verify that hardware elements behave as specified under fault injection.

3.4 Software Development (Part 6)

Part 6 specifies requirements for software development. Key requirements scale with ASIL level:

- **Software Architectural Design:** Must demonstrate freedom from interference between safety and non-safety software. Design notations and formal review are required at ASIL C/D.
- **Software Unit Design and Implementation:** Coding guidelines (e.g., MISRA C++) mandatory at all ASIL levels. Use of dynamic memory allocation and recursion is restricted or prohibited at higher ASIL levels.
- **Testing:** Statement coverage (ASIL A/B), branch coverage (ASIL B/C), MC/DC coverage (ASIL C/D). Back-to-back testing of model-based code against specification required at ASIL C/D.

- **Verification:** Formal methods, model checking, and static analysis (tools like Helix QAC, Polyspace) are recommended/required at ASIL C/D.

4. ISO 26262 Standard Parts Overview

ISO 26262:2018 (second edition) consists of 12 parts:

Part	Title	Key Content
Part 1	Vocabulary	Terms, definitions, abbreviations used throughout the standard
Part 2	Management of Functional Safety	Safety management, confirmation measures, safety culture
Part 3	Concept Phase	Item definition, HARA, safety goals, FSC, ASIL determination
Part 4	Product Dev: System Level	Technical safety requirements, system design, integration testing
Part 5	Product Dev: Hardware Level	Hardware architectural design, PMHF, hardware integration
Part 6	Product Dev: Software Level	SW architectural design, unit design, coding guidelines, unit testing
Part 7	Production & Operation	Production, operation, service, decommissioning requirements
Part 8	Supporting Processes	Configuration management, change management, qualification of tools
Part 9	ASIL-Oriented & Safety Analysis	ASIL decomposition, dependent failure analysis, safety analysis methods
Part 10	Guidelines	Non-normative guidance on applying the standard
Part 11	Guidelines on Application of ISO 26262 to Semiconductor Development	ASIL-relevant sections
Part 12	Adaptation for Motorcycles	Specific adaptations for two-wheeled road vehicles

5. Hazard Analysis and Risk Assessment (HARA) Methodology

HARA is the systematic process for identifying hazards and determining safety goals. It is the foundational activity of the concept phase and determines the ASIL for all safety requirements in the system.

5.1 HARA Process Steps

- **1. Situational Analysis:** Identify the operational situations in which the item operates (e.g., highway driving, low-speed urban, parking, wet road, night).
- **2. Hazard Identification:** For each function, identify what could go wrong — unintended activation, failure to activate, degraded performance, spurious output.
- **3. Hazardous Event Formation:** Combine each hazard with each operational situation to form hazardous events.
- **4. S/E/C Classification:** Assign severity, exposure, and controllability parameters.
- **5. ASIL Determination:** Apply the ASIL classification table.
- **6. Safety Goal Definition:** For each ASIL A–D hazardous event, define a safety goal expressed as a functional safety requirement (e.g., 'The ACC system shall not cause unintended braking exceeding 3 m/s²').

5.2 HARA Example: Adaptive Cruise Control (ACC)

For an ACC system, a representative HARA might identify the following hazardous events:

- **Unintended maximum deceleration on motorway (S3, E4, C2) → ASIL C:** ACC falsely brakes to maximum deceleration while following traffic at 130 km/h. High severity (rear-end collision risk), high

- exposure (motorway ACC use is frequent), partially controllable (driver can release brake) → ASIL C safety goal.
- **Failure to brake for stationary target (S3, E3, C3) → ASIL D:** ACC fails to detect and brake for a stationary vehicle. Life-threatening, occurs in high-speed situations, not controllable → ASIL D.
- **Unintended acceleration (S2, E4, C2) → ASIL B:** ACC accelerates when driver expects steady speed. Severe but usually controllable by driver braking → ASIL B.
- **Incorrect speed setpoint display (S0, E4, C0) → QM:** Display shows wrong target speed. No direct injury potential → QM only.

6. Safety Mechanisms and Architecture Patterns

Safety mechanisms are the technical measures implemented to prevent or detect faults and prevent them from causing safety violations. ISO 26262 requires that the safety architecture demonstrate sufficient coverage of identified failure modes through appropriate mechanisms.

6.1 Common Safety Mechanisms in ADAS

- **Plausibility checks:** Cross-validate sensor inputs against each other or expected physical ranges. Example: radar and camera target distance should agree within a tolerance; disagreement triggers degradation.
- **Watchdog timers:** Hardware or software timers that must be regularly reset by the main processing cycle. A missed reset indicates a CPU lockup or scheduling failure and triggers a safe state.
- **Signal degradation / health monitoring:** Monitor signal quality indicators, timeout counters, and CRC/checksum errors. Degrade functionality gracefully when signal health falls below thresholds. This pattern is central to ISO 26262 Part 4 safety architecture.
- **End-to-end (E2E) protection:** AUTOSAR E2E library provides CRC and sequence counter protection on safety-critical signals transmitted via CAN/Ethernet, detecting transmission errors and message loss.
- **Redundancy:** Dual-channel architectures where two independent processing paths must agree before a safety-critical actuation command is issued. Disagreement triggers a safe state.
- **Safe state definition:** Every ADAS function must define its safe state — the condition the system transitions to upon fault detection. For ACC, the safe state is typically deactivation with driver notification. For AEB, the safe state might be enabling emergency braking but disabling non-emergency braking.

6.2 Fail-Safe vs. Fail-Operational

Traditional automotive safety used fail-safe approaches (shut down on fault detection). Highly Automated Driving (HAD/L3+) increasingly requires fail-operational architectures where the system continues operating in a degraded mode for a sufficient time to allow the driver to take over (the Minimum Risk Condition). This significantly increases system complexity and ASIL requirements for the monitoring and degradation logic.

6.3 Freedom from Interference

ISO 26262 Part 6 requires that software of different ASIL levels (or QM software alongside ASIL software) demonstrate freedom from interference — that lower-integrity software cannot corrupt higher-integrity software through shared memory, timing interference, or communication channels. This is achieved through: memory protection units (MPU), time partitioning in AUTOSAR OS, and architectural separation of safety-relevant software components from QM components.

7. Application to Key ADAS Functions

7.1 Adaptive Cruise Control (ACC)

ACC maintains a set speed and time gap to a lead vehicle by controlling throttle and braking. Safety requirements typically span ASIL B (unintended acceleration, minor deceleration errors) to ASIL C (significant unintended deceleration or failure to brake in highway scenarios). Key safety goals: (1) ACC shall not cause unintended deceleration exceeding $X \text{ m/s}^2$ without driver intent; (2) ACC shall correctly release control and alert the driver upon sensor degradation; (3) ACC target selection shall not confuse stationary infrastructure (bridges, signs) with moving targets.

7.2 Automatic Emergency Braking (AEB)

AEB autonomously applies emergency braking to prevent or mitigate collisions. Failure modes include false activations (nuisance braking that can cause rear-end collisions) and missed activations (failure to protect

against genuine hazards). AEB typically carries ASIL D requirements for forward collision cases at highway speeds. The safety architecture must include: multi-sensor confirmation, plausibility checks against road topology, suppression logic for infrastructure objects, and E2E protected actuation commands.

7.3 Lane Keeping Assist (LKA) and Lane Departure Warning (LDW)

LKA applies steering torque to keep the vehicle within lane boundaries. Safety requirements are typically ASIL B–C. Key concerns: false activation causing sudden lane departure, failure during lane change, performance in poor visibility. Safety mechanisms: lane marking confidence thresholds, driver torque override detection, deactivation on turn signal.

7.4 Sensor Fusion Safety Considerations

Multi-sensor fusion (radar + camera + LiDAR) introduces dependencies between sensors. ISO 26262 Part 9 addresses dependent failure analysis — if a single cause (e.g., heavy rain, direct sunlight) degrades multiple sensors simultaneously, the fusion output may fail while individual channel diagnostics show 'healthy'. Safety architectures must demonstrate that common-cause failures are either prevented by sensor diversity or detected by cross-sensor plausibility monitoring.

8. Verification, Validation, and Testing

ISO 26262 distinguishes verification (are we building the product right?) from validation (are we building the right product?). Both are required at each level of the V-model.

8.1 Software Verification Methods by ASIL

- **ASIL A:** Statement coverage ($SC \geq 100\%$), code reviews, module testing.
- **ASIL B:** Branch coverage ($BC \geq 100\%$), walkthrough reviews, module testing, back-to-back testing.
- **ASIL C:** MC/DC coverage, formal reviews (inspection), back-to-back testing, semi-formal verification.
- **ASIL D:** MC/DC coverage, formal inspection, formal methods (where applicable), back-to-back testing, hardware-in-the-loop (HIL) validation.

8.2 Safety Analysis Methods

- **FMEA (Failure Mode and Effects Analysis):** Bottom-up analysis — for each component, enumerate all failure modes and trace their effects to the system level. Used for hardware ASIL analysis.
- **FTA (Fault Tree Analysis):** Top-down analysis — start from the safety goal violation and decompose the causes using AND/OR logic gates. Supports PMHF calculations.
- **FMEDA (Failure Modes, Effects and Diagnostic Analysis):** Extension of FMEA that quantifies diagnostic coverage of safety mechanisms. Required for Part 5 PMHF and SPFM/LFM calculations.
- **Dependent Failure Analysis (DFA):** Identifies common cause failures that could defeat redundancy. Required when ASIL decomposition is used.

8.3 Simulation-Based Testing

Simulation is increasingly used for ISO 26262 validation, particularly for ADAS functions where physical testing of all hazardous scenarios is impractical. Scenario-based testing using tools like ADTF, CARLA, CarMaker, or PreScan allows systematic coverage of the HARA scenarios. A key requirement is that simulation models are sufficiently validated against real-world behaviour to provide credible evidence for the safety case.

9. Key Terminology Reference

Term	Definition
ASIL	Automotive Safety Integrity Level. Risk classification A–D (D = highest) plus QM.
FSC	Functional Safety Concept. High-level safety measures derived from safety goals.
HARA	Hazard Analysis and Risk Assessment. Process to identify hazards and assign ASILs.
Safety Goal	Top-level safety requirement derived from HARA. Expressed as functional requirement, not implementation.
Item	System or combination of systems that implement a function at the vehicle level (e.g., the ACC system).
Element	System, software, hardware, or hardware part. Sub-unit of an item.
PMHF	Probabilistic Metric for random Hardware Failures. Quantitative reliability target (e.g., $< 10^{-8}/h$ for ASIL D).
SPFM	Single-Point Fault Metric. Fraction of single-point faults covered by safety mechanisms.
LFM	Latent Fault Metric. Fraction of latent faults that would not be perceived by the driver.
Safe State	Operating mode with acceptable risk, entered upon fault detection (e.g., ACC deactivation).
Controllability	Ability of driver or road user to avoid harm. C0 = always controllable; C3 = barely controllable.
E2E Protection	AUTOSAR End-to-End library. Adds CRC and sequence counters to safety-critical messages.

FMEA	Failure Mode and Effects Analysis. Bottom-up systematic analysis of component failure modes.
FTA	Fault Tree Analysis. Top-down deductive analysis starting from undesired top-level event.
DFA	Dependent Failure Analysis. Analysis of common-cause failures that defeat redundancy.
MISRA C++	Coding standard for safety-critical C++ software. Subset of C++ with defined rules to avoid undefined behavior.
Confirmation Measures	Safety review, audit, and assessment activities confirming compliance with ISO 26262.
SOTIF	Safety of the Intended Functionality (ISO/PAS 21448). Addresses hazards from correct but insufficient system behavior.

10. ISO 26262 and Machine Learning / AI Systems

ISO 26262 was designed for deterministic E/E systems. Applying it to machine learning components in ADAS (perception neural networks, behaviour prediction models) presents significant challenges that the industry is actively addressing through supplementary standards and guidelines.

- **The core problem:** ISO 26262 requires formal specification of requirements, deterministic behaviour, and structured test coverage. ML models learn behaviour from data rather than from formal specifications, making traditional V&V approaches insufficient.
- **ISO/TR 4804 and SOTIF (ISO 21448):** SOTIF specifically addresses scenarios where the system functions as intended but the intended function is insufficient (e.g., a pedestrian detection model that works correctly but misses partially-occluded pedestrians). SOTIF and ISO 26262 are complementary.
- **ASAM OpenX and scenario-based testing:** Systematic scenario coverage using OpenSCENARIO, OpenDRIVE, and OpenCRG formats is becoming the industry approach to demonstrating safety for ML-based perception.
- **Uncertainty quantification:** ML models must provide confidence/uncertainty estimates that feed into the ADAS safety architecture's plausibility checks and degradation logic.
- **Training data requirements:** ISO 26262 principles applied to ML require documentation of training data provenance, bias analysis, and out-of-distribution detection capability.

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